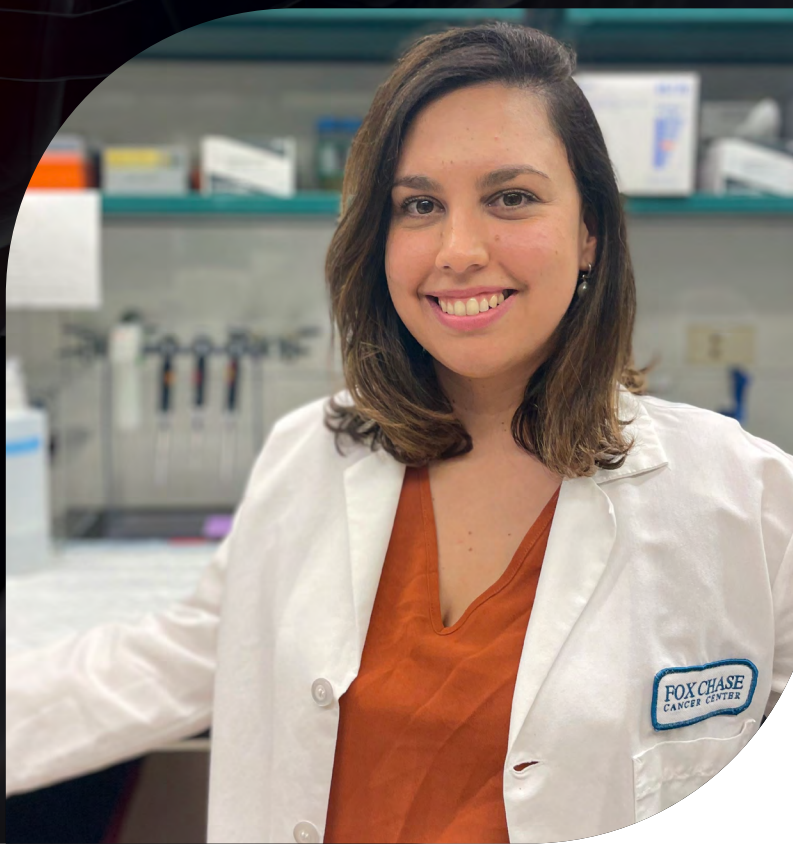
VIEW ONLINE

Fox Chase Cancer Center

Studying miRNA Signatures in Colorectal Cancer Cases

At Fox Chase Cancer Center, scientists like Mariana Fragoso, a postdoctoral associate in the Cancer Prevention and Control Program, are studying microRNA-based markers that can be used to detect colorectal cancer cases early — and potentially lead to better patient outcomes, especially among those at high risk of developing colorectal cancer.



MicroRNA Research at Fox Chase Cancer Center

Fragoso joined the cancer program after completing her Ph.D. in pathology at Sao Paulo State University in Brazil. She studies microRNA expression in ulcerative colitis-associated colorectal cancer cases. In her experiments, Fragoso works with preclinical mouse models, *in vitro* cancer cell lines, and patient tumor biopsies. “These models mimic the disease very well and are helpful for both understanding the underlying biology and identifying solutions that can ultimately be applied to patients,” she says.

For these studies, Fox Chase scientists work with microRNA expression data and RNA sequencing results, as well as protein information from immunohistochemistry and ELISA experiments. In one project, Fragoso analyzed formalin-fixed, paraffin-embedded (FFPE) biopsies collected from patients at various stages of ulcerative colitis. She used the Purigen Ionic® Purification System and the Ionic® FFPE to Pure RNA kits to extract RNA from her samples.

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>
> **Mariana Fragoso, Ph.D.**
> Post-doctoral Associate
> Cancer Prevention and Control Program

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The goal of this study was to identify a biomarker signature associated with the development of precancerous lesions in patients with ulcerative colitis. Over time, patients with this disease can develop two kinds of dysplasia: one with a flat morphology and one with an elevated growth component (mushroom-like). Although these high-risk individuals routinely undergo colonoscopies to identify tumors early in their development, flat lesions are much harder to detect and can be missed until it is too late. "We wanted to identify a microRNA signature that can be used to detect these lesions long before they become malignant, ideally from a blood sample," Fragoso says. "By studying samples from different times during the disease, we can see how the expression of microRNAs changes during the development of the disease."

Most scientists are familiar with the challenges of extracting RNA from FFPE samples. Besides low yields, such RNA is often contaminated or badly degraded, making downstream analysis difficult. In earlier projects, Fragoso frequently found herself struggling to get sufficient RNA for analysis from biopsies. And there was very little room for error given the amount of sample available.

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"We used to have a column-based kit for the extraction process, but the quality of the RNA and the yield from this approach was typically very poor," she says.

There were also a lot of steps involved in preparing the samples and scientists had to monitor the column during the experiment.

These are exactly the kinds of situations that the Purigen system and kits were designed to address. Fragoso was introduced to the Purigen technology by her lab supervisor. She tested the system on a handful of samples to see how it stacked up against the existing column-based method in the lab.

The difference between the two approaches was immediately clear, from the sample prep through to the final output.

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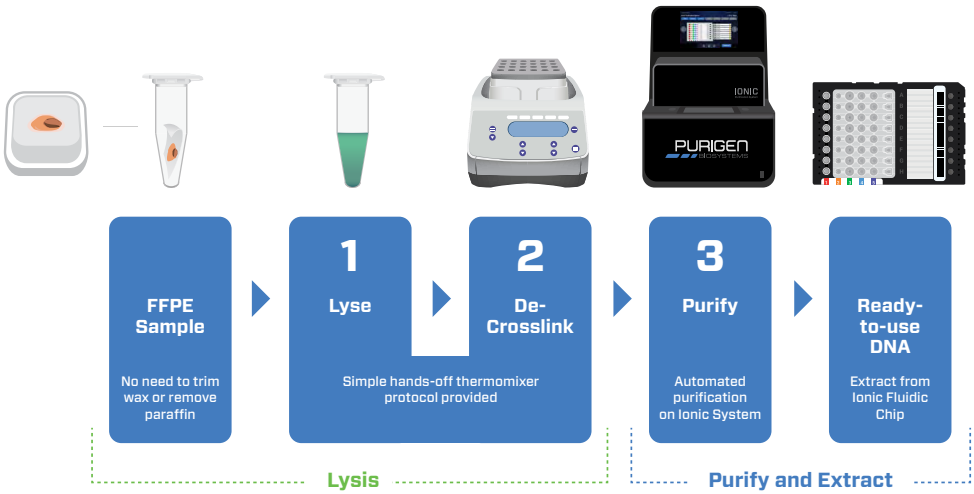
The Ionic Advantage and Solution

"Purigen's system has very simple protocols, is fast, and the quality of the RNA I got was amazing," Fragoso says. It also has fewer steps than the column-based approach. For example, there is no need for a deparaffinization step; the Purigen system takes care of this during the run. Once samples are loaded, scientists can work on other tasks while the machine runs.

The yield is also significantly better. From a 10-micron FFPE sample, Fragoso was able to extract about 17 nanograms of RNA per microliter using the Purigen system.

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Purigen RNA Extraction Workflow



"I need just about 10 nanograms of RNA for each RNA expression experiment that I do. With Purigen, I extracted as much as 850 nanograms from the samples," she says — more than enough to perform all the experiments she needed to do. Best of all, this performance came without breaking the bank. "The cost of the Purigen [method] is comparable to that of the column-based method," Fragoso says.

Access to sufficient RNA from FFPE samples is enabling Fragoso to make discoveries that could improve cancer detection and prevention in the future. In one study, she extracted RNA from biopsy samples from ulcerative colitis patients and analyzed microRNA expression. She successfully validated one of the microRNAs — miR-200a — in the proposed cancer detection signature. She also constructed a network of the genes targeted by all of the microRNAs in the signature and confirmed a prior prediction that when miR-200a is upregulated, it downregulates the expression of a gene that regulates methylation. Fragoso and her colleagues are now trying to understand the role of this interaction in colorectal cancer development among ulcerative colitis patients.

"For scientists like us who are working with FFPE samples and interested in studying RNA expression, Purigen is a very good option because of the high quality of the extracted RNA," Fragoso says. "Also, the technical support team is easy to work with and responsive to questions."

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